

medications and provide alerts for when to take medicine and automating refilling at the chemist. Whereas, other applications can monitor health parameters like blood pressure and share it with a doctor.

Connected equipment like X-ray machines can notify the monitoring system when an RFID tag worn by a patient is placed under the machine. IoMT identifiers can track the location of every doctor, nurse, patient and other medical equipment staff within a medical facility at any time. AI can predict which patient is most likely to be readmitted based on analyses over 100,000 data points. Predictive analytics can do other things like suggesting treatments, detecting diseases and identifying genetic factors affecting the patient. IoMT sensors can not only monitor but can immediately report when predefined criteria are exceeded.

3D training tools like 3D Medical help doctors learn about human anatomy without an actual dissection. These tools improve the doctors' understanding of patients by taking them through each stage of a disease. Tools like EchoPixel render patient-specific anatomy in an interactive VR format to increase knowledge, leading to faster and better care. EchoPixel takes data from CT and MRI scans, and transforms it into 3D holographic images to view and interact with patient tissues and organs as if these are real physical objects.

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The IoMT, the emergence of distance health

The IoMT refers to a connected system of medical devices and software applications. It includes remote monitoring of patient health and hospital asset management, followed by data analysis of applications for clinicians and patients. Everything from hospital beds to scanners can now be connect-

ed using technologies such as Bluetooth tags. Hospitals can track patients and treatments by accessing data via apps on their phones from connected devices like implantable glucose monitoring systems and others.

Wireless health monitoring. This offers a quality life for users and provides critical physical, biochemical and genomic data to healthcare professionals. Wireless and mobile cellular networking are increasing the demand for mobile-based solutions, both in general-ward hospital environments and home-care scenarios.

Wireless monitoring has shown unprecedented levels of analysis and understanding into disease progression, diagnosis and therapies. The idea is to allow patients to become responsible individuals in the matters of their own health.

This technology allows continuous, intelligent monitoring of multiple vital signs such as ECG heart rate, body temperature, respiration and activity level in real time, allowing earlier detection and prediction of adverse events such as heart attacks, falls or hypoglycaemia, with the help of appropriate external sensors, such as three-axis accelerometer, digital thermometers, pressure gauges, strain gauges and so on. This allows patients to monitor vital signs, be aware of what they should do in a distress

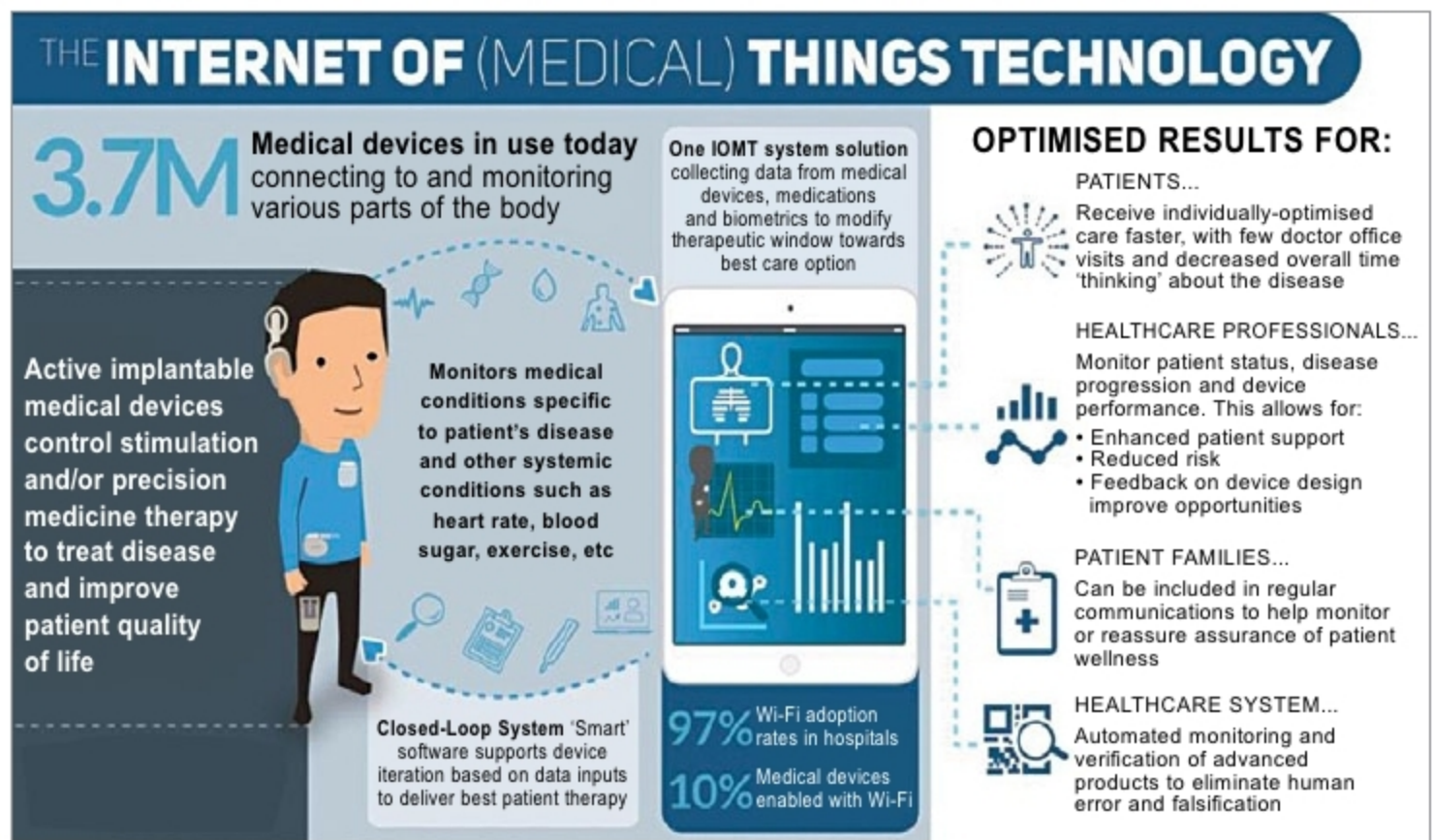
condition and learn to take preventive actions against chronic illnesses.

Benefits of health monitoring apps. Healthcare apps can be used to monitor health, track activity, adhere medication or notify care providers in case of a health emergency. Patients can immediately get e-consultations through apps like Push Doctor. This increases accessibility, particularly for patients who may be house-bound, bed-ridden or physically unable to go to a doctor.

Role of data analytics

Data managed by medical organisations includes medical insurance claims, hospital asset records, patient health records, pharmacies' medicine records, data collected from IoT devices or monitoring systems. This data helps in tracking patients and their progress. Healthcare systems can be transformed by deploying technologies like AI, machine learning, blockchain and cloud computing to restructure the way healthcare is delivered, measured and paid for.

The ability to interpret and analyse data inputs has the potential to transform many aspects of the healthcare industry. Key applications for Big Data in healthcare include the IoMT, predictive analysis and patient monitoring in real time. AI and machine learning provide clinical decision support,



The IoMT technology (Credit: Nexeon)